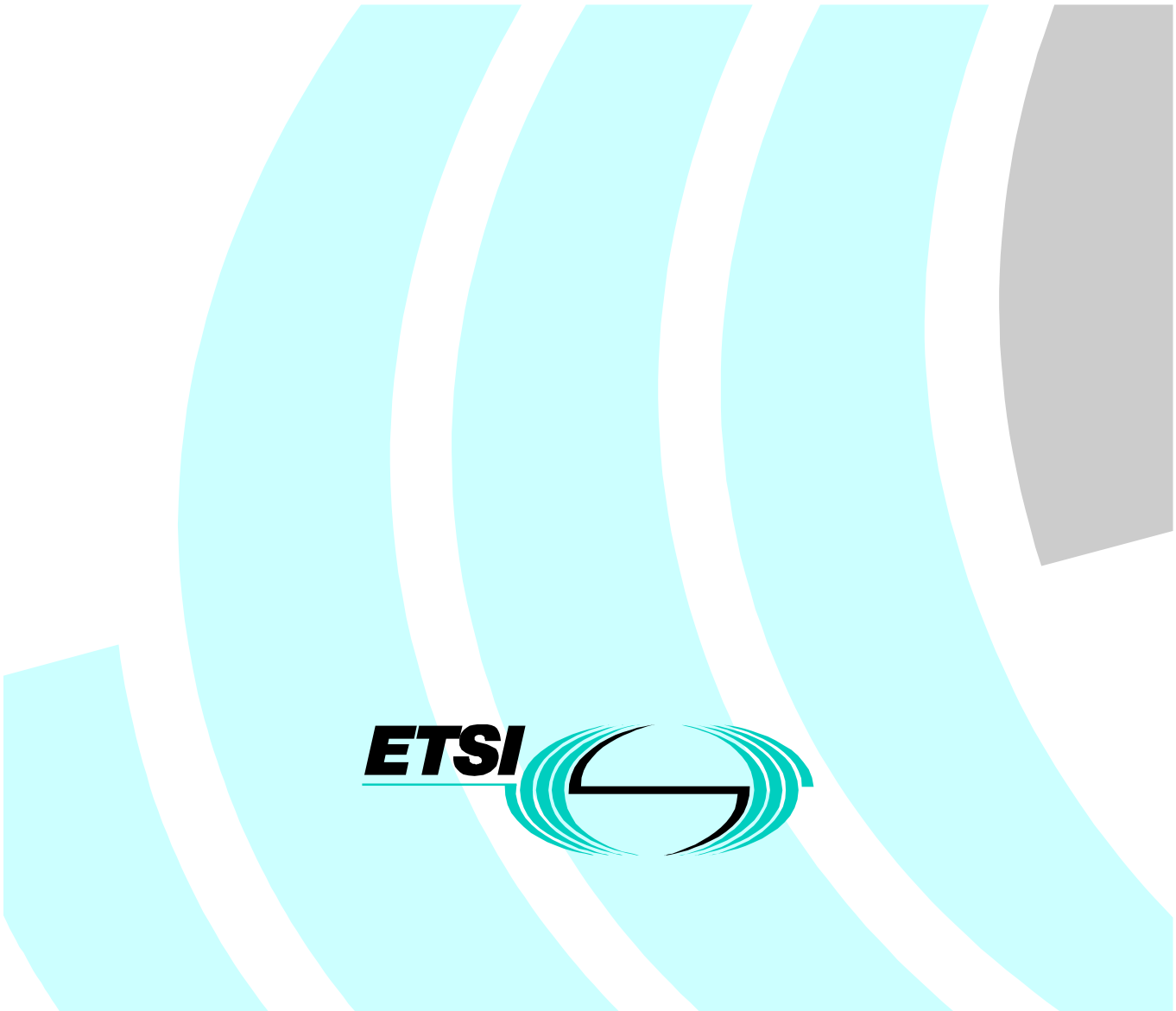
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Postal address
F-06921 Sophia Antipolis Cedex - FRANCE

Office address
650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16
Siret N° 348 623 562 00017 - NAF 742 C
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° 7803/88

Internet
secretariat@etsi.fr
Individual copies of this ETSI deliverable
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Contents

Intellectual Property Rights 4

Foreword 4

1 Scope 5

2 References 5

3 Environmental test specifications 5
3.1 Specification T 2.1: Very careful transportation 6
3.2 Specification T 2.2: Careful transportation 9
3.3 Specification T 2.3: Public transportation 12
3.4 Notes to tables 15
3.4.1 General note 15
3.4.2 Notes to tables 1 to 6 15

Bibliography 18

History 19

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Foreword

This European Standard (Telecommunications series) has been produced by ETSI Technical Committee Environmental Engineering (EE).

The present document consists of 2 parts as follows:

Part 1: "Classification of environmental conditions";

NOTE 1: Specifies different standardized environmental classes covering climatic and biological conditions, chemically and mechanically active substances and mechanical conditions during storage, transportation and in use.

Part 2: "Specification of environmental tests".

NOTE 2: Specifies the recommended test severities and test methods for the different environmental classes.

Each part of the standard is divided into sub-parts. Sub-part 2-0 forms a general overview of Part 2.

This sub-part 2-2 deals with transportation.

National transposition dates	
Date of adoption of this EN:	3 September 1999
Date of latest announcement of this EN (doa):	31 December 1999
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1 Scope

The present document specifies test severities and methods for verification of the required resistibility of equipment according to the relevant environmental class.

The tests defined in Part 2-2 of this multi-part EN apply to transportation of equipment covering the environmental conditions stated in ETS 300 019-1-2 [1].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.

A non-specific reference to an ETS shall also be taken to refer to later versions published as an EN with the same number.

[1] ETS 300 019-1-2: "Equipment Engineering (EE); Environmental conditions and environmental tests for telecommunications equipment; Part 1-2: Classification of environmental conditions; Transportation".

[2] IEC 60068-2: "Environmental testing - Part 2: Tests".

[3] ISO 4180-2 (1980): "Complete, filled transport packages - General rules for the compilation of performance test schedules - Part 2: Quantitative data".

[4] ETS 300 019-2-0: "Equipment Engineering (EE); Environmental conditions and environmental tests for telecommunications equipment; Part 2-0: Specification of environmental tests; Introduction".

3 Environmental test specifications

The detailed descriptions of the environmental conditions are given in clauses 4 and 5 of ETS 300 019-1-2 [1].

ETS 300 019-2-0 [4] forms a general overview of Part 2 of the present document.

If the equipment is normally transported in a packed state then it shall be tested in its packaging.

3.1 Specification T 2.1: Very careful transportation

Specification T 2.1 applies to transportation by air and by road on good quality road surfaces where special care has been taken with respect to low temperatures, handling and type of vehicle. See tables 1 and 2.

Table 1: Test specification T 2.1: Very careful transportation - climatic tests

Environmental parameter			Environmental Class 2.1	Environmental test specification T2.1: Very careful transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic Severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Air temperature	low	(°C)	-25	-25	6 h	IEC 60068-2-1	Ab: Cold	1
	high	unventilated (°C)	+70	+70	6 h	IEC 60068-2-2	Bb: Dry heat	
		ventilated or outdoors (°C)	+40	none				
	change	air/air (°C) (°C/min)	-25/+30	-25/+30 1,0	5 cycles t1 = 3h	IEC 60068-2-14	Nb: Change of temperature	3a
		air/water (°C)	+40/+5	none				**3b**
Humidity	relative	slow temperature (%) change (°C)	95 +40	93 +30	4 d	IEC 60068-2-56	Cb: Damp heat steady state	4
		rapid temperature (%) change (°C)	95 -25/+30	90-100 +40	2 cycles	IEC 60068-2-30	Db: Damp heat cyclic Variant 1	5
	absolute	rapid temperature (°C) change (g/m^3)	+70/+15 60	none				
Air	pressure	low (kPa)	70	none				6
		change	no					
	speed	(m/s)	20	none				7
Water	rain	intensity (mm/min)	6	none				8
		low temperature (°C)	no					
	other sources	(m/s)	1					7
	wetness		wet surfaces					9
Radiation	solar	(W/m^2)	1 120					10
	heat	(W/m^2)	600					10

Environmental parameter			Environmental Class 2.1	Environmental test specification T2.1: Very careful transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic Severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Chemically active substances	sulphur	SO_2 (mg/m^3)	1,0	none				11
		H_2S (mg/m^3)	0,5	none				11
	chlorine	salt	sea and road salt mist	none				11
		Cl_2 (mg/m^3)	no					11
		HCl (mg/m^3)	0,5	none				11
	nitrogen	NO_x (mg/m^3)	1,0	none				11
		NH_3 (mg/m^3)	3,0	none				11
	hydrogen fluoride HF	(mg/m^3)	0,03	none				11
	ozone O_3	(mg/m^3)	0,1	none				11
Mechanically active substances	dust	sedimentation ($mg/(m^2h)$)	3,0	none				12
		suspension (mg/m^3)	no					
	sand	(mg/m^3)	100	none				12
Flora and fauna	micro organisms		mould, fungus, etc.	none				13
	rodents, insects		rodents, etc.	none				13

NOTE 1: no = this condition does not occur in this class.
NOTE 2: none = verification is required only in special cases.
NOTE 3: n = number of note, see subclause 3.4.

Table 2: Test specification T 2.1: Very careful transportation - mechanical tests

Environmental parameter			Environmental Class 2.1	Environmental test specification T 2.1 Very careful transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic Severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Vibration	sinusoidal	displacement (mm) acceleration (m/s^2) frequency range (Hz) axes of vibration	3,5 10 2-9 9-200 15 200-500	none				14
	random	ASD (m^2/s^3) (dB/oct) frequency range (Hz) axes of vibration	1 0,3 10-200 200-2 000	1,0 -3 5-20 20-200 3	3 x 30 minutes	IEC 60068-2-64	Fh: Vibration, broad-band random (digital control)	15
Shocks	shocks	shock spectrum duration (ms) acceleration (m/s^2) mass (kg) number of bumps direction of bumps	no					
Fall	free fall	height (mm) mass (kg) attitude	no					
	toppling around	mass (kg) edges	no					
Acceleration	steady state	(m/s^2)	20	none				19
Load	static load	(kPa)	5	none				20
Miscellaneous	rolling and pitching	angle (deg) period (s)	no					

NOTE 1: no = this condition does not occur in this class.
NOTE 2: none = verification is required only in special cases.
NOTE 3: n = number of note, see subclause 3.4.

3.2 Specification T 2.2: Careful transportation

Specification T 2.2 applies to transportation by air, by road on good quality road surfaces, by ship and by train with specially designed shock-reducing buffers and where special care has been taken with respect to low temperatures and handling. See tables 3, 4 and 7.

Table 3: Test specification T 2.2: Careful transportation - climatic tests

Environmental parameter			Environmental Class 2.2	Environmental test specification T2.2: Careful transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Air temperature	low	(°C)	-25	-25	72 h	IEC 60068-2-1	Ab: Cold	1
	high	unventilated (°C)	+70	+70	72 h	IEC 60068-2-2	Bb: Dry heat	
		ventilated or outdoors (°C)	+40	none				
	change	air/air (°C) (°C/min)	-25/+30	-25/+30 1,0	5 cycles t1 = 3h	IEC 60068-2-14	Nb: Change of temperature	3a
		air/water (°C)	+40/+5	none				3b
Humidity	relative	slow temperature (%) change (°C)	95 +40	93 +40	4 d	IEC 60068-2-56	Cb: Damp heat steady state	4
		rapid temperature (%) change (°C)	95 -25/+30	90-100 +40	2 cycles	IEC 60068-2-30	Db: Damp heat cyclic Variant 1	
	absolute	rapid temperature (°C) change (g/m^3)	+70/+15 60	none				5
Air	pressure	low (kPa)	70	none				6
		change	no					
	speed	(m/s)	20	none				7
Water	rain	intensity (mm/min)	6	none				8
		low temperature (°C)	no					
	other sources	(m/s)	1					7
	wetness		wet surfaces					9
Radiation	solar	(W/m^2)	1 120					10
	heat	(W/m^2)	600					10

Environmental parameter			Environmental Class 2.2	Environmental test specification T2.2: Careful transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Chemically active substances	sulphur	SO_2 (mg/m^3)	1,0	none				11
		H_2S (mg/m^3)	0,5	none				11
	chlorine	Salts	sea and road salt mist	none				11
		Cl_2 (mg/m^3)	no					
		HCl (mg/m^3)	0,5	none				11
	nitrogen	NO_x (mg/m^3)	1,0	none				11
		NH_3 (mg/m^3)	3,0	none				11
	hydrogen fluoride HF	(mg/m^3)	0,03	none				11
	ozone O_3	(mg/m^3)	0,1	none				11
Mechanically active substances	dust	sedimentation ($mg/(m^2h)$)	3,0	none				12
		suspension (mg/m^3)	no					
	sand	(mg/m^3)	100	none				12
Flora and Fauna	micro organisms		mould, fungus, etc.	none				13
	rodents, insects		rodents, etc.	none				13

NOTE 1: no = this condition does not occur in this class.
NOTE 2: none = verification is required only in special cases.
NOTE 3: n = number of note, see subclause 3.4.

Table 4: Test specification T 2.2: Careful transportation - mechanical tests

Environmental parameter			Environmental Class 2.2	Environmental test specification T 2.2: Careful transportation				
	Parameter	Detail parameter	Characteristic severity	Test severity	Duration	Reference	Method	Notes
Vibration	sinusoidal	displacement (mm) acceleration (m/s^2) frequency range (Hz) axes of vibration	3,5 10 2-9 9-200 200-500	none				14
	random	ASD (m^2/s^3) (dB/oct) frequency range (Hz) axes of vibration	1,0 0,3 10-200 200-2 000	1,0 -3 5-20 20-200 3	3 x 30 minutes	IEC 60068-2-64	Fh: Vibration, broad-band random (digital control)	15
Shocks	shocks	shock spectrum duration (ms) acceleration (m/s^2) mass (kg) number of bumps direction of bumps	Type I 11 100	half sine 6 11 100 50 ≤ 50 > 50 6	100 in each direction	IEC 60068-2-29	Eb: Bump	16
Fall	free fall	height (m) mass (kg) attitude	0,25 0,25 0,1 < 20 20-100 > 100	see table 7		IEC 60068-2-32	Ed: Procedure 1	17
	toppling around	mass (kg) edges	< 20 20-100 > 100 any no no	none				18
Acceleration	steady state	(m/s^2)	20	none				19
Load	static load	(kPa)	5	none				20
Miscellaneous	rolling and pitching	angle (deg) period (s)	no					

NOTE 1: no = this condition does not occur in this class.
NOTE 2: none = verification is required only in special cases.
NOTE 3: n = number of note, see subclause 3.4.

3.3 Specification T 2.3: Public transportation

Specification T 2.3 applies to transportation by air, by road on all qualities of road surface, by ship and by train and where some care has been taken with respect to low temperatures. See tables 5 to 7.

Table 5: Test specification T 2.3: Public transportation - climatic tests

Environmental parameter			Environmental Class 2.3	Environmental test specification T2.3: Public transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Air temperature	low	(°C)	-40	-40	72 h	IEC 60068-2-1	Ab: Cold	1
	high	unventilated (°C)	+70	+70 (and +85)	72 h (6 h)	IEC 60068-2-2	Bb: Dry heat	2
		ventilated or outdoors (°C)	+40	none				
	change	air/air (°C) (°C/min)	-40/+30	-40/+30 1,0	5 cycles t1 = 3 h	IEC 60068-2-14	Nb: Change of temperature	3a
		air/water (°C)	+40/+5	none				3b
Humidity	relative	slow temperature (%) change (°C)	95 +45	93 +40	4 d	IEC 60068-2-56	Cb: Damp heat steady state	4
		rapid temperature (%) change (°C)	95 -40/+30	90-100 +40	2 cycles	IEC 60068-2-30	Db: Damp heat cyclic Variant 1	
	absolute	rapid temperature (°C) change (g/m^3)	+70/+15 60	none				5
Air	pressure	low (kPa)	70	none				6
		Change	no					
	speed	(m/s)	20	none				7
Water	rain	intensity	6 mm/min	0,01 m^3/min, 90 kPa	3 min/m^2 or 15 min	IEC 60068-2-18	Rb: Impacting water, Method 2.2	8
		low temperature (°C)	no					
	other sources	(m/s)	1					7
	wetness		wet surfaces					9
Radiation	solar	(W/m^2)	1 120					10
	heat	(W/m^2)	600					10

Environmental parameter			Environmental Class 2.3	Environmental test specification T2.3: Public transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Chemically active substances	sulphur	SO_2 (mg/m^3)	1,0	none				11
		H_2S (mg/m^3)	0,5	none				11
	chlorine	salts	sea and road salt mist	none				11
		Cl_2 (mg/m^3)	no					11
		HCl (mg/m^3)	0,5	none				11
	nitrogen	NO_x (mg/m^3)	1,0	none				11
		NH_3 (mg/m^3)	3,0	none				11
	hydrogen fluoride HF	(mg/m^3)	0,03	none				11
	ozone O_3	(mg/m^3)	0,1	none				11
Mechanically active substances	dust	sedimentation ($mg/(m^2h)$)	3,0	none				12
		suspension (mg/m^3)	no					
	sand	(mg/m^3)	100	none				12
Flora and fauna	micro organisms		mould, fungus, etc.	none				13
	rodents, insects		rodents,etc.	none				13

NOTE 1: no = this condition does not occur in this class.
NOTE 2: none = verification is required only in special cases.
NOTE 3: n = number of note, see subclause 3.4.

Table 6: Test specification T 2.3: Public transportation - mechanical tests

Environmental parameter			Environmental Class 2.3	Environmental test specification T 2.3: Public transportation				
Type	**Parameter**	**Detail parameter**	**Characteristic severity**	**Test severity**	**Duration**	**Reference**	**Method**	**Notes**
Vibration	sinusoidal	displacement (mm) acceleration (m/s^2) frequency range (Hz) axes of vibration	3,5 10 2-9 9-200 15 200-500	none				14
	random	ASD (m^2/s^3) (dB/oct) frequency range (Hz) axes of vibration	1 0,3 10-200 200-2 000	1,0 -3 5-20 20-200 3	3 x 30 minutes	IEC 60068-2-64	Fh: Vibration, broad-band random (digital control)	15
Shocks	shocks	shock spectrum duration (ms) acceleration (m/s^2) mass (kg) number of bumps direction of bumps	Type I 11 100 Type II 6 300	half sine 6 180 ≤ 50 11 100 > 50 6	100 in each direction	IEC 60068-2-29	Eb: Bump	16
Fall	free fall	height (m) mass (kg) attitude	1,2 1,0 0,25 < 20 20-100 > 100	see table 7		IEC 60068-2-32	Ed: Procedure 1	17
	toppling around	mass (kg) edges	< 20 20-100 > 100 any any any	none				18
Acceleration	steady state	(m/s^2)	20	none				19
Load	static load	(kPa)	10	none				20
Miscellaneous	rolling and pitching	angle (deg) period (s)	±35 8	none				

NOTE 1: none = verification is required only in special cases.
NOTE 2: n = number of note, see subclause 3.4.

Table 7: Free fall test severities for test specifications T 2.2 and T 2.3

	Free fall test height [m]	
Mass [kg]	**Class T 2.2 (note)**	**Class T 2.3 (note)**
< 10	0,8	1,0
< 15	0,6	1,0
< 20	0,6	0,8
< 30	0,5	0,6
< 40	0,4	0,5
< 50	0,3	0,4
< 100	0,2	0,3
> 100	0,1	0,1
NOTE 1: Duration: 1 fall on each face. If the mass is > 5 kg and if the normal attitude is specified, 2 falls in the specified attitude only.		
NOTE 2: Values specified in ISO 4180-2 [3].		

3.4 Notes to tables

3.4.1 General note

The present document applies to transportation of equipment covering environmental conditions stated in ETS 300 019-1-2 [1]. The notes have been added to explain the main reasons for recommended tests or to explain why no test has been recommended even if there is a characteristic severity given.

The equipment should be tested in the state in which it is normally transported if possible. For example, if the equipment is in a packed state, then it should be tested in its packaging. If the equipment is transported both with and without its packaging it may be necessary to perform tests for both configurations. For some tests and equipment, the test may be more severe for the packaged rather than the unpacked equipment.

3.4.2 Notes to tables 1 to 6

NOTE 1: (Air temperature, low)

- If the equipment is tested without any package or if the equipment is small the duration may be decreased.

NOTE 2: (Air temperature, high)

- An additional test at 85°C for 6 h shall be conducted on unpacked equipment only. The additional test includes solar radiation effects.

NOTE 3: (Air temperature, change)

- 3a) (air/air)

- The change of temperature test is normally used to check design tolerancing and the range is not important. However in this class condensation may occur. The lowest recommended test values of IEC 60068-2-14 [2] Test Nb have been chosen. For unpacked equipment with a mass < 5 kg test Na is applied.

- 3b) (air/water)

- The effect of rapid temperature change experienced by the equipment when it rains on a warm day is considered to be less severe than those experienced during the change of temperature (air/air; Test Nb) and therefore no additional test is needed.

NOTE 4: (Humidity, relative, slow temperature change)

- Test required for unpacked equipment only.

NOTE 5: (Humidity, absolute, rapid temperature change)

- Condensation is included in IEC 60068-2-30 [2] Test Db and temperature change is partly included in IEC 60068-2-14 [2] Test Nb.

NOTE 6: (Air pressure, low)

- For normal applications where the effect of air pressure is evaluated at the component level therefore no test is recommended.

NOTE 7:

- There is no IEC 60068-2 [2] test for this parameter.

NOTE 8: (Water, rain)

- The water test may be omitted in tables 2.1 and 2.2 because in these classes the equipment will be exposed to rain only for short duration. IEC 60068-2-18 [2] Test Rb method 2.2 has been chosen even if it does not imitate normal rain. It is a simple hand held shower test, easy to perform and can demonstrate that the specimen design is adequately toleranced to survive this condition.

NOTE 9: (Water, other sources, wet surfaces)

- If the equipment is in contact with wet surface the corrosion effect and degeneration effect has to be considered.

NOTE 10: (Radiation, solar, heat)

- The effect of direct sun radiation is included in the higher test value in IEC 60068-2-2 [2] Test Bb, as described in note 2. Photochemical tests can be made separately for components and materials.

NOTE 11: (Chemically active substances)

- For chemically active substances the characteristic severity should be considered when choosing components and materials. No test is recommended in this standard. Characteristic severities of chemically active substances are max. values.

NOTE 12: (Mechanically active substances)

- For mechanical substances the packaging is supposed to protect the equipment against dust and sand, therefore no test is recommended. The Ievels of dust both sedimentation and suspension are far lower than the lowest severity recommended in IEC 60068-2-68 [2] Test Lb.

NOTE 13: (Flora, fauna)

- The characteristic severity should be considered when choosing component and materials.

NOTE 14: (Vibration, sinusoidal)

- Random vibration is considered to be a more realistic test for this condition, therefore no sinusoidal test is recommended. The severities are given as peak values.

NOTE 15: (Vibration, random)

- The most energy is in low frequencies and therefore the most realistic test has been described with a:

- -3 dB/oct slope from 20 Hz to 200 Hz;

- ASD (Acceleration Spectral Density) vibrations are of greatest significance in the vertical direction. If normal attitude during transportation is specified, then the severity for the horizontal axes ASD is reduced by a factor 10.

- Acceleration RMS (for information only): 8,76 m/s^2.

- 2,77 m/s^2, when the test severity is reduced by a factor 10.

NOTE 16: (Shocks)

- During transportation, the number of shocks is expected to be high, so the bump test is more adequate for testing. The severities are given as peak values. For masses > 500 Kg no bump test is required. Test severity values not specified in IEC 60068-2 [2].

- The specified test severities for m < 50 kg and m > 50 kg have been chosen to have the same energy per mass unit for both situations. Bumps are of greatest significance in the vertical direction. If normal attitude during transportation is specified, 100 bumps have to be applied along that direction only.

NOTE 17: (Free Fall)

- ISO severities are chosen because they are considered to be more realistic.

NOTE 18: (Toppling around)

- No test is required because the effect is included in IEC 60068-2-32 [2] Test Ed, free fall.

NOTE 19: (Acceleration, steady state)

- This characteristic severity is considered to be insignificant and therefore no test is recommended.

NOTE 20: (Load)

- This condition is for packaged equipment only. Packaging should be designed with this requirement in mind.

Bibliography

The following material, though not specifically referenced in the body of the present document (or not publicly available), gives supporting information.

- ETR 035: "Equipment Engineering (EE); Environmental engineering; Guidance and terminology".
- IEC 60068-1: "Environmental testing - Part 1: General and guidance".

History

Document history		
Edition 1	May 1994	Publication as ETS 300 019-2-2
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